Multiplying Bigger Numbers

Three-digit times two-digit: six pieces, or two rows.

243 × 36: six pieces

	200	40	3
30	6,000	1,200	90
6	1,200	240	18

Hundreds, tens and ones across the top; tens and ones down the side.

6,000 + 1,200 + 90 + 1,200 + 240 + 18 = 8,748

The same in two rows

	243
×	36
1,458	× 6
7,290	× 30
8,748	

Row 1: 243 × 6 = 1,458 (the bottom pieces).
Row 2: 243 × 30 = 7,290 (the top pieces).
243 × 30 is 243 × 3 tens = 729 tens.
Add the two rows: 8,748.

Your turn!

312 × 23 = 7,176

For teachers

Goal: multiply three- and four-digit numbers by two-digit numbers (5.NBT.5). The two rows are the area model's pieces added in groups. Say “243 times 3 tens is 729 tens”, never “put down a zero”.

Multiplying Bigger Numbers: Practice

Split, multiply each piece, add.

Part 1 Hundreds times tens

3 hundreds × 4 tens = 12 thousands.

1)	300 × 40 = 12,000	2)	200 × 60 = 12,000	3)	500 × 70 = 35,000
4)	400 × 30 = 12,000	5)	700 × 20 = 14,000	6)	600 × 90 = 54,000
7)	800 × 50 = 40,000	8)	900 × 40 = 36,000	9)	200 × 80 = 16,000
10)	300 × 70 = 21,000	11)	500 × 50 = 25,000	12)	400 × 60 = 24,000

Multiplying Bigger Numbers: Practice (continued)

Part 2 Six pieces The parts are on the outside. Fill in each piece.

13) **234 × 15 = 3,510**

	200	30	4
10	2,000	300	40
5	1,000	150	20

2,000 + 300 + 40 + 1,000 + 150 + 20 = 3,510

14) **162 × 37 = 5,994**

	100	60	2
30	3,000	1,800	60
7	700	420	14

3,000 + 1,800 + 60 + 700 + 420 + 14 = 5,994

15) **425 × 23 = 9,775**

	400	20	5
20	8,000	400	100
3	1,200	60	15

8,000 + 400 + 100 + 1,200 + 60 + 15 = 9,775

Multiplying Bigger Numbers: Practice (continued)

Part 3 Split it yourself Write the parts on the outside, then fill in each piece.

16) **318 × 24 = 7,632**

	300	10	8
20	6,000	200	160
4	1,200	40	32

6,000 + 200 + 160 + 1,200 + 40 + 32 = 7,632

17) **256 × 43 = 11,008**

	200	50	6
40	8,000	2,000	240
3	600	150	18

8,000 + 2,000 + 240 + 600 + 150 + 18 = 11,008

18) **173 × 52 = 8,996**

	100	70	3
50	5,000	3,500	150
2	200	140	6

5,000 + 3,500 + 150 + 200 + 140 + 6 = 8,996

Multiplying Bigger Numbers: Practice (continued)

Part 4 Two rows Row 1: times the ones. Row 2: times the tens. Then add.

19) 214 × 32
428 × 2
6,420 × 30
6,848

20) 365 × 24
1,460 × 4
7,300 × 20
8,760

21) 128 × 57
896 × 7
6,400 × 50
7,296

22) 432 × 18
3,456 × 8
4,320 × 10
7,776

23) 251 × 46
1,506 × 6
10,040 × 40
11,546

24) 617 × 35
3,085 × 5
18,510 × 30
21,595

25) 183 × 29
1,647 × 9
3,660 × 20
5,307

26) 526 × 14
2,104 × 4
5,260 × 10
7,364

27) 342 × 67
2,394 × 7
20,520 × 60
22,914

Part 5 Multiply No help this time.

28) 326 × 21 = 6,846

29) 147 × 38 = 5,586

Multiplying Bigger Numbers: Practice (continued)

Part 5 Multiply (continued)

30) 512 × 46 = 23,552

31) 238 × 17 = 4,046

32) 403 × 25 = 10,075

33) 671 × 34 = 22,814

34) 295 × 42 = 12,390

35) 184 × 63 = 11,592

36) 736 × 15 = 11,040

37) 459 × 27 = 12,393

38) 128 × 74 = 9,472

39) 387 × 56 = 21,672

40) 604 × 39 = 23,556

41) 215 × 88 = 18,920

Multiplying Bigger Numbers: Practice (continued)

Part 5 Multiply (continued)

42) 548 × 23 = 12,604

43) 963 × 12 = 11,556

44) 372 × 45 = 16,740

45) 819 × 26 = 21,294

46) 256 × 91 = 23,296

47) 437 × 68 = 29,716

Part 6 Four digits times one digit No help this time.

48) 2,134 × 3 = 6,402

49) 4,206 × 7 = 29,442

50) 1,573 × 4 = 6,292

51) 3,058 × 6 = 18,348

52) 6,125 × 8 = 49,000

53) 2,947 × 5 = 14,735

Multiplying Bigger Numbers: Practice (continued)

Part 6 Four digits times one digit (continued)

54) 5,312 × 9 = 47,808

55) 1,806 × 4 = 7,224

56) 7,243 × 3 = 21,729

57) 3,615 × 6 = 21,690

Part 7 Multiply

58) 672 × 35 = 23,520

59) 763 × 43 = 32,809

60) 728 × 41 = 29,848

61) 163 × 54 = 8,802

62) 353 × 47 = 16,591

63) 182 × 39 = 7,098

64) 672 × 59 = 39,648

65) 236 × 54 = 12,744

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

66) 526 × 16 = 8,416

67) 162 × 15 = 2,430

68) 621 × 38 = 23,598

69) 113 × 29 = 3,277

70) 873 × 19 = 16,587

71) 113 × 22 = 2,486

72) 194 × 12 = 2,328

73) 664 × 59 = 39,176

74) 153 × 33 = 5,049

75) 542 × 49 = 26,558

76) 371 × 26 = 9,646

77) 736 × 48 = 35,328

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

78) 358 × 17 = 6,086

79) 848 × 59 = 50,032

80) 237 × 38 = 9,006

81) 275 × 42 = 11,550

82) 344 × 58 = 19,952

83) 355 × 48 = 17,040

84) 148 × 17 = 2,516

85) 464 × 12 = 5,568

86) 497 × 24 = 11,928

87) 527 × 55 = 28,985

88) 459 × 14 = 6,426

89) 604 × 24 = 14,496

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

90) $182 \times 36 =$ 6,552

91) $883 \times 43 =$ 37,969

92) $121 \times 13 =$ 1,573

93) $165 \times 47 =$ 7,755

Part 8 Remember? Problems from earlier packets.

94) $3.6 + 2.75 =$ 6.35

95) $8.2 - 4.67 =$ 3.53

96) 4.28 to the nearest tenth = 4.3

97) 0.45 > 0.405

98) in 3.142 the 4 is worth 0.04

99) $2.7 \times 1{,}000 =$ 2,700

Multiplying Bigger Numbers: Practice (continued)

Part 9 Word problems Split, multiply, add.

100) A school bus holds 48 kids. How many kids can 125 buses hold?

Number sentence 125 × 48 = 6,000

Answer 6,000 kids

Draw or show your work

101) A farm packs 236 boxes of eggs. Each box has 12 eggs. How many eggs is that?

Number sentence 236 × 12 = 2,832

Answer 2,832 eggs

Draw or show your work

102) A plane flies 465 miles each day for 21 days. How far does it fly?

Number sentence 465 × 21 = 9,765

Answer 9,765 miles

Draw or show your work

103) A box of paper has 1,250 sheets. How many sheets are in 8 boxes?

Number sentence 1,250 × 8 = 10,000

Answer 10,000 sheets

Draw or show your work

104) Challenge: A theater has 34 rows of 28 seats. 875 seats are full. How many are empty?

Number sentence 34 × 28 = 952, 952 − 875 = 77

Answer 77 seats

Draw or show your work

Check-In: Multiplying Bigger Numbers

Show what you know. No help this time.

Mastery: 8 of 10 (80%)

Part 1 Hundreds times tens

1) 300 × 90 = 27,000

2) 800 × 90 = 72,000

3) 700 × 70 = 49,000

Part 2 Two rows

4)

418 × 23

1,254 × 3

8,360 × 20

9,614

5)

152 × 47

1,064 × 7

6,080 × 40

7,144

6)

324 × 26

1,944 × 6

6,480 × 20

8,424

Part 3 Multiply

7) 382 × 19 = 7,258

8) 527 × 34 = 17,918

9) 243 × 17 = 4,131

10) 275 × 36 = 9,900